

STAT 112

INTRODUCTION TO STATISTICS AND PROBABILITY II

SESSION 4- Bernoulli Random Variable & Distribution

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Session Outline

- 1 Bernoulli Random Variable
- 2 Bernoulli Distribution Function
- 3 Moments of Bernoulli R.V.
- 4 Worked Examples

Session Overview

This session introduces students to Bernoulli probability distribution. The experiment that gives rise to the Bernoulli random variable and properties of the distribution will be discussed. Calculation of probabilities using the Bernoulli mass function and theoretical derivation of the mean and variance of the Bernoulli random variable will be presented.

Learning Outcomes

At the end of the session, students will be able to:

- 1 Explain and describes the experiment leading to Bernoulli random variable.
- 2 State the probability mass function of the Bernoulli distribution and show that it is a probability function.
- 3 Prove the expectation and variance of Bernoulli distribution.
- 4 Compute probabilities using Bernoulli distribution function.

Definition of Bernoulli random variable

- The Bernoulli random variable is also known as $(0, 1)$ random variable or indicator random variable.
- The random variable takes the value of 0 or 1.
- A Bernoulli trial is one of the simplest experiments you can conduct in probability and statistics.
- It is an experiment where one can have one of two possible outcomes.

Experiment of Bernoulli random variable

- Suppose there are two possible outcome on a trial of an experiment.
- The two possible outcomes are designated as "Success" and the other "Failure".
- One can only occur at a time, i.e. if success occurs, failure will not occur.
- Let A be the event of a "Success", then $\bar{A}$ is the event of a "Failure".
- Suppose that the probability of a success is p , where $0 < p < 1$.
 $P(A) = P(\text{Success}) = p$ and $P(\bar{A}) = P(\text{Failure}) = 1 - p = q$,
where $p + q = 1$.
- Let X represent the number of Successes in a Bernoulli trial, then X can assume values 0 and 1.

Bernoulli Distribution

- X will take a value 0 when a Failure occurs and X will take the value 1 when a Success occurs.

That is: $X = 0$, if and only if a Failure Occur and
 $X = 1$, if and only if a Success Occur.

- Now:

$$P(X = 0) = P(\text{Failure}) = q \quad (1)$$

$$P(X = 1) = P(\text{Success}) = p \quad (2)$$

- Combining eqns. (1) & (2) we have

$$P(X = x) = p^x q^{1-x}, \quad x = 0, 1$$

Probability Mass Function of Bernoulli

The Bernoulli probability mass function is given by:

Bernoulli PMF

$$P(X = x) = p^x q^{1-x}, \quad x = 0, 1$$

where $0 < p < 1$ and $q = 1 - p$.

- The above function is known as the Bernoulli distribution.
- It is denoted by $X \sim \text{bern}(p)$, that is; X has a Bernoulli distribution with parameter p .

Proof of Legitimacy Condition

- (a.) $P(X = x) \geq 0 \forall x$. Given that

$$P(X = x)p^x q^{1-x},$$

Since $0 < p < 1$, it implies that $p^x > 0 \forall x$, and $q^{1-x} \geq 0 \forall x$

$$\therefore P(X = x) \geq 0, \forall x.$$

- (b.) $\sum P(X = x) = 1$

$$\begin{aligned}\sum_{x=0}^1 P(X = x) &= \sum_{x=0}^1 p^x q^{1-x} = p^0 q^{1-0} + p^1 q^{1-1} \\ &= q + p = 1 - p + p = 1\end{aligned}$$

- Hence $P(X = x)$ is a legitimate probability function.

Characteristics of the Bernoulli R.V.

Theorem 1

Given that $X \sim \text{bern}(p)$, then

$$E(X) = p \quad \text{and} \quad \text{Var}(X) = pq$$

Proof of Expectation

$$\begin{aligned} E(X) &= \sum_{\text{all } x} xP(X = x) = \sum_{x=0}^1 xp^xq^{1-x} \\ &= 0 \times p^0q^{1-0} + 1 \times p^1q^{1-1} \\ &= 0 + p = p \end{aligned}$$

Characteristics of the Bernoulli R.V.

Proof of Variance

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = E(X^2) - p^2$$

$$E(X^2) = \sum_{\text{all } x} x^2 P(X = x) = \sum_{x=0}^1 x^2 p^x q^{1-x}$$

$$= 0 \times p^0 q^{1-0} + 1^2 \times p^1 q^{1-1} = p$$

$$\therefore \text{Var}(X) = p - p^2 = p(1 - p)$$

$$= pq \quad \square$$

Example 1

In tossing a fair die, let X be a random variable defined as

$$X = \begin{cases} 0, & \text{when a 'six' does not occur} \\ 1, & \text{when a 'six' does occur} \end{cases}$$

- a Write down the probability mass function of X indicating the values of the random variable X assumes.
- b Find the mean and standard deviation of X .

Solution

a. PMF

$$P(X = 0) = P(\text{a six does not occur})$$

$$P(X = 1) = P(\text{a six does occur}) = \frac{1}{6}$$

$$P(X = 0) = 1 - P(X = 1) = \frac{5}{6}$$

$$\therefore P(X = x) = \left(\frac{1}{6}\right)^x \left(\frac{5}{6}\right)^{1-x}, \quad x = 0, 1$$

b. Standard deviation

$$\text{If } X \sim \text{bern}(p), \text{ then } \text{Var}(X) = pq = \frac{1}{6} \times \frac{5}{6} = \frac{5}{36}$$

$$\text{Standard deviation} = \sqrt{\text{Var}(X)} = \frac{\sqrt{5}}{6}$$

THANK YOU!!!